The Ayushman Bharat Digital Mission serves as the backbone for India's healthcare transformation, enabling providers to craft solutions through interoperability, unique health IDs, standardised registries and unified interfaces. Building on this foundation, disease surveillance apps, like Aarogya Setu and CoWIN, can be deployed to map non-communicable disease risks in vulnerable populations through ASHA workers. This digital infrastructure can enable big data analytics to drive prevention, which is crucial for managing India's growing disease burden. The collected data can further power AI tools to identify India-specific risk factors, fuelling research for population-specific treatments while guiding resource deployment and expansion of the care network.

IMPROVING ACCESS

To utilise the set-up fully, healthcare enablers can scale solutions like e-ICUs and digital operation theatres for stronger last-mile delivery. This will be crucial for rural areas, home to 65 per cent of the population but only 40 per cent of the hospitals, where interconnected platforms can enhance virtual healthcare delivery. Such digital solutions can help address shortages of specialist doctors while improving access, especially in Tier 2 and 3 cities where bed ratios lag at 0.5 per 1,000 people compared to 1.5 in Tier 1 cities. The impact is already visible in programmes like tuberculosis screening, where AI-powered imaging tools overcome radiologist shortages in remote areas, accelerating diagnosis and treatment initiation. This convergence of precision medicine with AI marks the next frontier in preventive healthcare, vital to managing India's growing non-communicable disease challenge.

By combining genetic testing, phenomics (the study of an organism's phenotype, or traits, across its genome) and AI, precision medicine enables creation of comprehensive 'health *patris*'—personalised health forecasts that empower doctors and patients to turn family health history into a proactive tool for disease prevention. Understanding genetic predispositions and their interaction with environmental factors allows healthcare providers to prescribe targeted interventions, including lifestyle modifications, preventive medications and screening protocols, to help patients stay disease-free for longer.

TRANSFORMING HEALTHCARE

The impact of this approach is particularly significant for high-burden conditions like cardiovascular disease and cancer. By identifying diseases in the pre-disease stage, precision medicine enables timely identification and intervention, preventing acute events like heart attacks and detecting cancers when they are most treatable. With cancer cases projected to rise, predictive precision medicine can offer both improved outcomes and more manageable healthcare costs. Advanced analytics can even guide population-level interventions, creating a more efficient and empathetic healthcare system.

THE CONVERGENCE OF PRECISION MEDICINE WITH AI MARKS **THE NEXT FRONTIER IN PREVENTIVE HEALTHCARE,** VITAL TO MANAGING INDIA'S NON-COMMUNICABLE DISEASE CHALLENGE

While technology and precision medicine provide powerful tools for healthcare transformation, their impact depends on skilled professionals who can effectively use these innovations. As India expands its network of medical colleges and research centres, aligning with the country's evolving disease patterns becomes important to create a future-ready workforce in addition to meeting global standards for the number of caregivers.

Standardisation of care is also vital for scaling quality healthcare. A majority of India's hospitals are small to medium outfits and only a small percentage is accredited by the National Accreditation Board for Hospitals & Healthcare Providers (NABH). The adoption of digital health standards by smaller hospitals is an important stepping stone toward NABH-grade care. When facilities integrate standardised digital protocols and systems, they take systematic steps toward better quality benchmarks, creating a pathway to accreditation while improving patient care.

THE PATH FORWARD

Large healthcare providers can catalyse transformation by collaborating with small and medium-sized hospitals via through structured skill enhancement workshops, protocol sharing and quality improvement initiatives. While this may not immediately lead to NABH accreditation for smaller facilities, it will create a foundation for standardised care delivery crucial to achieve the scale India needs.

While government-healthcare innovator partnerships drive systemic change, citizen engagement in preventive health is equally crucial. As Indians increasingly embrace digital health tools and preventive care, they become active participants in building a healthier nation. This shift from passive recipients to active participants in healthcare will be key to achieving Universal Health Coverage and 'Viksit Bharat', making 2025 a pivotal year in India's journey toward healthcare excellence. ■

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